# ASSEM MERZOUG

Automation & Systems Engineering — Internship

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## PROFILE

Electrical engineering graduate, now studying Automation and Systems Engineering at the University of Palermo. My work sits where control systems meet software: modelling a physical system, scripting the measurement and the report, and testing the result until the assumptions show themselves.

## SKILLS

**Systems and control.** Three-phase inverter modelling, Space Vector PWM, simulation of power electronics.

**Programming.** Python for modelling, data handling and tooling; Bash and PowerShell for environment automation.

**Environments.** Linux and Windows administration, Vagrant and VirtualBox for reproducible test machines, Git for version control.

**Testing.** Turning a physical or software system into a set of checks that either pass or produce evidence of why they did not.

## SELECTED WORK

### Three-phase inverter —SVPWM

Bachelor's final project: modelling and simulating a three-phase voltage inverter using Space Vector PWM, including the modulation logic and the simulated output waveforms.

### pdfsmith —a report engine in plain Python

A zero-dependency library that turns structured findings into a finished PDF. Demonstrates the same habit the role needs: take a repetitive engineering output and automate it properly.

### Active Directory lab as code

A Windows domain defined in a Vagrantfile with PowerShell provisioning —infrastructure written down and rebuilt on demand rather than configured by hand.

## EXPERIENCE

### Freelance —self-employed · 2021 to present

Five years of self-directed client work while completing the BSc, security-focused since 2023: agreeing the requirement, building it, and documenting it well enough to hand over.

## EDUCATION

### MSc Automation & Systems Engineering —University of Palermo

In progress.

### BSc Electrical Engineering

Final project: three-phase inverter with Space Vector PWM.

## CERTIFICATIONS IN PROGRESS

Security certification work (BSCP, CWES) alongside the degree; OSCP as the long-term target.